

Sushmitha Gorrepati

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PROFESSIONAL SUMMARY:

Highly skilled AI/ML Engineer and Full Stack Developer with around 5+ years of experience designing, developing, and deploying **enterprise-grade AI and software solutions** across banking, healthcare, and IT services. Proven expertise in **machine learning model development, LLM fine-tuning, and RAG pipelines** for NLP, summarization, and predictive analytics. Skilled in **microservices architecture and API development (REST/gRPC)** with strong backend (Python, Java, Spring Boot) and frontend (React.js, Redux) capabilities. Experienced in building **end-to-end MLOps workflows** using Azure, AWS, and GCP, with containerization (Docker, Kubernetes) and CI/CD automation (Jenkins, GitLab CI/CD, Azure DevOps). Adept in **database optimization** with PostgreSQL, MongoDB, and vector databases, ensuring scalability, compliance, and high performance in production systems. Collaborative in **Agile/Scrum** environments with experience of delivering AI solutions that reduce operational costs, improve decision-making, and enhance customer experience in large-scale enterprises.

EDUCATION:

Texas Tech Univeristy

Aug 2023 – May 2025

Masters of Science in Computer and Information Science **CGPA: 3.91/4.00**

TECHNICAL SKILLS:

- **Programming Languages:** Python, SQL (Advanced), Java (8/17), JavaScript (ES6+), TypeScript
 - **Frameworks & Libraries:** Spring Boot, Spring MVC, Spring Data, Spring Security, scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers, TFX, LangChain, FastAPI, Flask, React.js, Redux Toolkit, LangGraph
 - **Frontend Development:** React.js, Redux Toolkit, HTML5, CSS3, Material UI, Tailwind CSS, Bootstrap
 - **Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, Cassandra, Redis, ChromaDB, FAISS, Milvus
 - **Cloud Platforms & DevOps:** Azure (Azure Machine Learning, AKS, Data Factory, Synapse, Databricks, Application Insights, Azure Monitor); AWS (SageMaker, S3, Glue, Redshift, EMR, Lambda); GCP (BigQuery, Dataflow, Vertex AI, Cloud Run); Docker, Kubernetes, Jenkins, GitLab CI/CD, Azure DevOps
 - **Microservices & Messaging:** RESTful APIs, gRPC, GraphQL, Apache Kafka, Microservices Architecture
 - **Version Control & Collaboration:** Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Agile/Scrum
 - **Microservices & Messaging:** Microservices Architecture, RESTful APIs, gRPC, Apache Kafka
 - **Testing & QA:** JUnit, Mockito, TestNG, Selenium, PyTest
 - **Web Services & APIs:** REST, gRPC, GraphQL, JSON, XML
 - **Performance Optimization & Monitoring:** MLflow, Kubeflow, Azure Monitor, Cloud Logging, Cloud Monitoring, Log Analytics
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PROFESSIONAL EXPERIENCE

Client: Bank of America, Charlotte NC

Aug 2024-Present

Role: AI Software Engineer

Responsibilities:

- Developed and deployed ML models for NLP and time-series analysis using Python, scikit-learn, Keras, and Hugging Face Transformers, delivering production-grade AI services for real-time call transcription, summarization, and agent assistance.
- Built and optimized ML pipelines leveraging Random Forest, Logistic Regression, LSTMs, and Transformer-based models to improve accuracy on structured and unstructured call data, applying hyperparameter tuning and model ensembling.
- Fine-tuned LLMs (BERT, GPT, LLaMA) for summarization, sentiment analysis, and Q&A, improving accuracy by 20% and reducing inference latency with quantization and distillation techniques.
- Implemented low-latency semantic search by integrating LLaMA with ChromaDB in RAG pipelines, increasing contextual query relevance by 40% and enabling knowledge-grounded call insights.
- Engineered modular AI workflows with LangChain and enforced schema validation to ensure reliable chaining of multi-step LLM tasks, reducing production errors.
- Designed and developed RESTful and gRPC APIs using FastAPI for secure, high-throughput model inference and seamless integration with enterprise contact center platforms.

- Deployed containerized inference APIs on Azure Machine Learning with AKS orchestration, scaling dynamically to handle thousands of concurrent calls and improving throughput by 50%.
- Automated training, testing, and deployment workflows with Jenkins, GitLab CI/CD, and Azure DevOps, reducing release cycles by 30% while enforcing governance and security compliance.
- Optimized hybrid storage using PostgreSQL for call metadata and ChromaDB for vector embeddings, improving parallel query handling by 45% in live environments.
- Monitored and debugged AI services with Azure Monitor, Log Analytics, and Application Insights, ensuring 99.9% SLA compliance, detecting model drift, and triggering automated retraining.
- Collaborated with data scientists, MLOps, and compliance teams to integrate AI into enterprise workflows, aligning with banking regulations and operational requirements.

Environment: Python, SQL, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, FastAPI, REST/gRPC APIs, RAG (ChromaDB), PostgreSQL, Azure Machine Learning, Azure Kubernetes Service (AKS), MLflow, Jenkins, GitLab CI/CD, Azure DevOps, Docker, Kubernetes, JWT, RBAC, Agile/Scrum.

Client: Tata Consultancy Services Limited, Bangalore, India

Mar2022-Aug 2023

Role: ML Engineer

Responsibilities:

- Designed and delivered a secure, cloud-based healthcare analytics platform processing 10M+ medical records, integrating Python ML models with a Java Spring Boot microservices backend and React.js + Redux Toolkit frontend to deliver real-time insights for internal analytics teams.
- Built NLP pipelines in Python using TensorFlow, Keras, and scikit-learn for automating clinical note classification, patient sentiment analysis, and discharge summary generation, improving turnaround time for medical reviews.
- Fine-tuned transformer models (BERT, T5, RoBERTa) via Hugging Face Transformers, improving classification accuracy by 25% on domain-specific datasets such as EHR (Electronic Health Records) and claims data.
- Developed automated CI/CD pipelines for ML workflows using TFX and Kubeflow, enabling smooth model updates without interrupting existing workflows.
- Deployed ML workloads on AWS SageMaker for training and inference, used AWS S3 for HIPAA-compliant data storage, and leveraged AWS EMR for large-scale batch processing.
- Implemented semantic search and RAG pipelines using FAISS and Milvus to quickly retrieve relevant patient records and medical literature for care management teams.
- Integrated real-time data feeds from Tecton and Databricks Feature Store, enabling live model scoring for readmission risk prediction and population health analysis.
- Developed Java Spring Boot microservices for ingesting clinical and claims data, orchestrating ML inference via Python FastAPI endpoints, and serving results to analytics dashboards.
- Processed and transformed large datasets using Apache Spark and SparkSQL, reducing ETL latency by 35% and improving feature extraction efficiency.
- Built interactive dashboards in React.js + Redux Toolkit to display KPIs, patient risk scores, and predictive insights, ensuring ease of use for non-technical healthcare staff.
- Ensured compliance with HIPAA and internal security protocols by implementing RBAC, encryption at rest and in transit, and audit logging across all services.
- Collaborated with domain experts, data teams, and compliance specialists in Agile/Scrum sprints to ensure ML solutions aligned with healthcare workflows, data quality standards, and regulatory requirements.

Environment: Python, Java, SQL, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, FastAPI, Spring Boot, React.js, Redux, Apache Spark, SparkSQL, PostgreSQL, FAISS, Milvus, RAG, TFX, Kubeflow, Tecton, Databricks Feature Store, MLflow, Docker, Kubernetes, AWS (SageMaker, S3, EMR).

Client: Hexaware Technologies, Hyderabad India

Jan2020-Feb 2022

Role: ML/Java Developer

Responsibilities:

- Developed and maintained enterprise-grade Java applications for backend services and APIs, supporting mission-critical client operations.
- Designed and integrated machine learning models using Python (Scikit-learn, TensorFlow) to automate document classification, anomaly detection, and customer behavior analysis.

- Built RESTful APIs and microservices using Java (Spring Boot) to expose ML functionalities and streamline integration with front-end and legacy systems.
- Collaborated with data scientists and business analysts to gather requirements, preprocess data, and validate model outcomes using real-world scenarios.
- Implemented NLP-based solutions for extracting insights from unstructured text data, including customer feedback, support tickets, and chat transcripts.
- Developed automated data ingestion and preprocessing pipelines using Python and Java for structured and unstructured datasets from multiple data sources.
- Optimized application performance by tuning JVM parameters, using caching (Redis), and implementing multi-threaded processing for high-throughput systems.
- Participated in Agile development cycles, performing code reviews, writing unit/integration tests (JUnit/TestNG), and using Git, JIRA, Jenkins for CI/CD.
- Integrated AI-driven components into existing Java-based systems, ensuring smooth model deployment, version control, and rollback mechanisms.
- Coordinated with QA teams to conduct functional, regression, and performance testing of AI-enhanced Java applications.
- Developed scalable predictive and streaming analytics using Apache Spark MLlib, Flink, and Kafka, enhancing real-time AI pipelines
- Implemented MLOps practices using MLflow, Kubeflow, Docker, Kubernetes, and Jenkins CI/CD, ensuring reliable model training, deployment, and monitoring.
- Supported production-ready AI deployments by developing cloud-native AI solutions with GCP Vertex AI

Environment: Java, Spring Boot, Python, Scikit-learn, TensorFlow, NLP, REST APIs, Jenkins, Git, JIRA, Docker, MongoDB, MySQL

PROJECTS

Agentic AI for Customer Support

- Designed a multi-agent customer support system using Python, LangChain, LangGraph, and Qdrant, enabling intelligent handling of travel queries through RAG pipelines, semantic search, and agent orchestration with observability via LangSmith on AWS.

Natural Language to SQL Interface

- Developed an AI-powered Text-to-SQL assistant using GPT-3.5, ChromaDB, and Streamlit to enable natural language database querying with real-time execution and interactive visualizations.